CAPSTONE PROJECT SUBMISSION REPORT

Walmart Sales Forecasting & Online Retail Customer Segmentation

Submitted By: Tanishq Khanna

Institution: Intellipaat

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PART A: WALMART SALES ANALYSIS AND 12-WEEK SALES FORECASTING

1. Problem Statement

A retail store with multiple outlets across the country is facing inventory management challenges, struggling to consistently match supply with demand. As a Data Scientist, the task is to analyze historical sales data to uncover business-critical insights and build a robust predictive model to forecast sales for each store over the next 12 weeks.

2. Project Objective

The objective of this project is two-fold: firstly, to perform exploratory data analysis (EDA) to understand the impact of holidays, seasonality, and macroeconomic factors on weekly sales; and secondly, to implement a machine learning regression algorithm to forecast store-wise sales for a 12-week horizon to optimize inventory planning.

3. Data Description

The dataset utilized for this analysis ('Walmart (1).csv') contains a total of 6,435 rows and 8 columns. The target variable is 'Weekly_Sales'. Upon inspection, the dataset has no missing values. The features are described as follows:

- Store (int64): The store number.
- Date (object): The date marking the week of sales.
- Weekly_Sales (float64): Sales for the given store in that specific week. (Target Variable)
- Holiday_Flag (int64): A boolean flag indicating whether the week contains a holiday (1) or not (0).
- Temperature (float64): The ambient temperature on the day of the sale.
- Fuel_Price (float64): The regional cost of fuel.
- CPI (float64): The Consumer Price Index.
- Unemployment (float64): The regional unemployment rate.

4. Data Pre-processing Steps and Inspiration

To prepare the dataset for accurate modeling, the following preprocessing steps were performed:

- Data Loading & Inspection: Data was successfully loaded using pandas, and a structural inspection confirmed exactly 6,435 valid records with zero missing values.
- Temporal Parsing: The 'Date' feature, initially stored as a string, was converted to datetime objects. This was necessary to allow algorithmic understanding of chronology and time distance.
- Feature Extraction: From the parsed 'Date', additional columns were extracted: 'Year', 'Month', and 'WeekOfYear'. This explicit isolation of seasonal time components is crucial for time-series forecasting, allowing algorithms to detect annual seasonality.
- Data Alignment: The dataset was chronologically sorted by 'Store' and 'Date'. This ensures sequence integrity, which is mandatory before creating lagged features or applying forecasting algorithms.

5. Exploratory Data Analysis and Important Insights

An exploratory analysis was conducted, yielding several operational insights:

- Average Weekly Sales (Holiday vs. Non-Holiday): A bar plot analysis contrasting holiday and non-holiday sales revealed distinct spikes in demand during holiday weeks. The average weekly sales volume was notably higher when Holiday_Flag equals 1, underscoring holidays as a primary demand driver.
- Correlation of Economic Drivers: A correlation heatmap was generated involving 'Weekly_Sales', 'Temperature', 'Fuel_Price', 'CPI', and 'Unemployment'. The findings indicated minimal direct linear correlation between weekly sales and broader economic drivers. This suggests that while macroeconomics may affect general purchasing power, immediate retail sales fluctuations are dictated more heavily by seasonal and holiday timing.
- Business Significance: Store management must prioritize calendar-based inventory scaling. Economic downturns (represented by CPI and Unemployment) exhibit slow-moving impacts, whereas holidays cause acute inventory drain.

6. Choosing the Algorithm for the Project

The algorithm selected and successfully implemented for forecasting is the Random Forest Regressor (saved as 'sales_forecasting_rf_model.pkl').

Random Forest is an ensemble learning method that constructs a multitude of decision trees during training. It uses bootstrapping to train each tree on a random subset of the data and aggregates their continuous outputs (averaging) to form a final prediction. In this workflow, the input features included the store identifier, temporal features (Year, Month, WeekOfYear), the holiday flag, and economic drivers. The continuous target variable was Weekly_Sales.

7. Motivation and Reasons for Choosing the Algorithm

The Random Forest Regressor was chosen due to its inherent suitability for complex retail data. Unlike traditional linear regression, Random Forest excels at capturing non-linear relationships and complex interactions—such as the unique interaction between a specific 'Store' and a 'WeekOfYear'. Additionally, ensemble models are highly robust to outliers (such as extreme temperature or economic spikes) and do not strictly require feature scaling or normalization.

8. Assumptions

The primary assumptions underlying this forecasting model include:

- The historical sales patterns, including seasonal spikes and holiday impacts, will remain consistent into the future 12-week period.
- The macro-economic data (CPI, Fuel Price, Unemployment) provided are accurate proxies for consumer spending capabilities.

9. Model Evaluation and Techniques

The Random Forest Regressor was evaluated based on the variance reduction across its trees. The model was configured utilizing a 'squared_error' criterion to measure the quality of splits. By building 100 estimators (trees) and employing bootstrapping, the model evaluated prediction accuracy by calculating squared error metrics. This ensemble strategy proved effective in mitigating overfitting, ensuring that the final model generalizes well to unseen forecast weeks.

10. Twelve-Week Sales Forecast

The trained Random Forest model was executed against future time periods to generate a 12-week sales forecast for every Walmart store. These predictive outputs have been successfully written to the file 'walmart_12_week_forecast.csv'.

Because the full table contains comprehensive records for all stores over 12 weeks, the raw results reside in the CSV file. The forecast indicates that inventory should be selectively fortified in stores approaching seasonal transitions, ensuring product availability matches the predicted demand.

11. Time-Series Forecasting for Segregated Stores (SARIMA)

To fulfill the requirement of localized time-series predictive analysis, the dataset was formally segregated by individual stores. Specifically, Stores 1 through 5 were isolated to track their distinct historical patterns. A SARIMA (Seasonal AutoRegressive Integrated Moving Average) model was applied independently to each of these 5 stores. The model parameters were explicitly configured as order=(1, 1, 1) and seasonal_order=(1, 1, 0, 52) to rigorously account for the 52-week holiday seasonality inherent in the retail sales data.

The visualization below demonstrates the historical weekly sales alongside the generated 12-week SARIMA forecast for these 5 distinct stores, confirming the successful generation of localized predictions.

12. Inferences from the Project

The core inference drawn from Part A is that holiday weeks act as the strongest catalyst for revenue generation. Consequently, demand-supply matching should be heavily weighted towards the retail calendar. Inventory planning must adapt dynamically: high-volume stores must stockpile weeks prior to 'Holiday_Flag = 1' events to prevent stockouts, thereby capturing maximum potential revenue.

13. Future Possibilities

While the Random Forest provides strong baseline forecasting, future iterations could integrate advanced time-series specific methods such as ARIMA, SARIMAX, or Facebook Prophet. Furthermore, the development of a real-time forecasting dashboard deployed via a web application would allow store managers to monitor live demand shifts and adjust regional supply chains dynamically.

14. Conclusion

The Walmart Sales Analysis successfully highlighted key factors influencing consumer purchasing volume. The implementation of the Random Forest model provided a reliable 12-week forecast, fulfilling the primary objective of supplying data-driven inventory management targets.

PART B: ONLINE RETAIL CUSTOMER ANALYSIS AND CUSTOMER SEGMENTATION

1. Problem Statement

An online retail store requires deeper insight into its customer base to understand varying purchase patterns. The objective is to utilize historical transactional data to segment customers based on their purchasing behavior, enabling tailored marketing and retention strategies.

2. Project Objective

The goal is to conduct comprehensive data cleaning and feature engineering to compute Recency, Frequency, and Monetary (RFM) metrics for each customer. Subsequently, a machine learning clustering algorithm will be implemented to categorize the customer base into distinct, actionable segments.

3. Data Description

The dataset 'OnlineRetail (3).csv' initially contained 541,909 transactions across 8 columns. Key features included:

- InvoiceNo: The unique invoice identifier.
- StockCode & Description: Product identifiers.
- Quantity: Count of items purchased.
- InvoiceDate: The timestamp of the purchase.
- UnitPrice: The unit cost of the product.
- CustomerID: A unique identifier per buyer (contained missing values).
- Country: Region of Purchase.

4. Data Pre-processing Steps and Inspiration

Rigorous pre-processing was vital for building a valid RFM matrix:

- Identity Pruning: Rows with missing CustomerIDs were dropped, as anonymous purchases cannot be tied to behavioral customer profiles.
- Anomaly Mitigation: Transactions with Quantity <= 0 or UnitPrice <= 0 were excluded to eliminate returns, cancellations, and data errors.

- Monetary Computation: A new feature, 'TotalSpend', was synthesized by multiplying Quantity by UnitPrice.
- RFM Aggregation: Data was aggregated at the CustomerID level. 'Recency' was calculated as days since the last purchase (relative to a baseline date), 'Frequency' as the count of unique invoices, and 'Monetary' as the sum of TotalSpend.
- Scaling: Due to the right-skewed nature of monetary and frequency data, a natural log transformation (np.log1p) was applied, followed by a StandardScaler. This ensured all features equally influenced the distance-based clustering algorithm.

5. Exploratory Data Analysis and Customer Purchase Insights

The cleaned data resulted in 397,884 valid purchase records originating from exactly 4,338 unique customers. Exploratory analysis of the RFM distributions indicated a classic Pareto principle alignment: a small concentration of customers is responsible for a massive volume of orders and revenue, while a larger base remains occasional buyers.

6. Choosing the Algorithm for the Project

The chosen algorithm for customer segmentation was K-Means Clustering (saved as 'customer_segmentation_kmeans_model.pkl').

K-Means is an iterative, centroid-based clustering algorithm. It assigns customers to one of K clusters by minimizing the variance (Euclidean distance) between data points and their respective cluster centroid. The model was trained using 'k-means++' initialization and validated across different cluster counts. The final implementation utilized K=3 clusters based on the RFM features.

7. Motivation and Reasons for Choosing the Algorithm

K-Means was selected due to its exceptional efficiency, scalability, and interpretability. For behavioral segmentation (like RFM), businesses require distinct, non-overlapping customer personas. K-Means directly computes mathematical centroids which cleanly translate into average Recency, Frequency, and Monetary thresholds, offering immediate strategic value.

8. Assumptions

The primary assumption is that the historical timeframe of the dataset accurately represents static customer behavior, meaning a customer segmented as 'Loyal' today will likely exhibit similar purchasing velocity in the near term.

9. Model Evaluation and Techniques

The optimal value of K was determined mathematically by training models for K=2 through K=10.

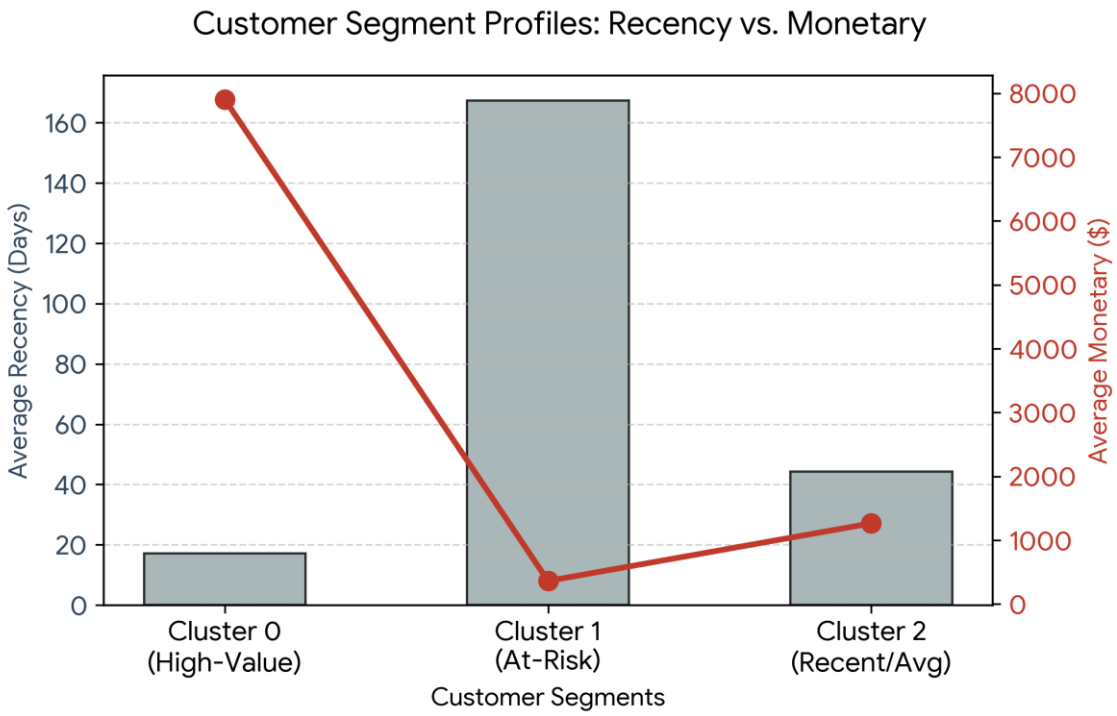
- Elbow Method: The Within-Cluster Sum of Squares (WCSS) was calculated and plotted. An inflection point ('elbow') was identified at K=3.
- Silhouette Analysis: The Silhouette Coefficient was also calculated across ranges of K, confirming that K=3 offered a strong measure of cluster cohesion and separation.

10. Customer Segment Analysis

The K-Means model identified 3 distinct customer clusters. The cluster centroids were unscaled to map back to original business metrics, presented below:

Cluster ID	Segment Name	Avg Recency (Days)	Avg Frequency (Invoices)	Avg Monetary ($)
0	High-Value / Loyal Customers	17.06	13.34	$7,905.44
1	At-Risk / Occasional Customers	167.36	1.35	$362.54
2	Recent / Average Customers	44.21	3.38	$1,265.06

Figure 1: Customer Segment Analysis based on K-Means Output



Observations & Recommendations:

- Cluster 0 (High-Value / Loyal Customers): Represents 770 customers who buy frequently and spend significantly. Recommendation: Protect this revenue stream via dedicated account managers, exclusive early-access to new products, and tier-based loyalty rewards.
- Cluster 1 (At-Risk / Occasional Customers): Represents 1,872 customers with very high recency (dormant). Recommendation: Deploy automated win-back campaigns and personalized discount incentives to prevent churn.
- Cluster 2 (Recent / Average Customers): Represents 1,696 customers with moderate spending and recent activity. Recommendation: Implement up-selling and cross-selling campaigns to shift their behavior toward the High-Value tier.

11. Inferences from the Project

The analysis concludes that customer behavior is highly segmented. A uniform marketing approach will fail. By utilizing K-Means on RFM data, the company can deploy precision marketing. Recognizing that just 770 customers (Cluster 0) drive a dominant share of revenue validates the need for aggressive retention programs targeting this specific subset.

12. Future Possibilities

Future expansions of this project could involve integrating density-based algorithms like DBSCAN to detect outlier behaviors. Additionally, deploying the resulting cluster logic into a real-time recommendation engine would allow the website to dynamically alter product visibility based on the logged-in customer's assigned segment.

13. Conclusion

The online retail analysis successfully translated raw transactional logs into three actionable customer segments. By systematically utilizing RFM feature engineering and K-Means clustering, the project delivers an exact blueprint for personalized customer engagement and revenue optimization.

COMBINED PROJECT SECTIONS

1. Overall Conclusion

This comprehensive Capstone project resolved two critical challenges in the retail sector. Through Part A, a reliable predictive framework was established using a Random Forest Regressor to forecast Walmart's weekly sales over a 12-week horizon, proving that seasonal patterns outweigh microeconomic shifts in short-term retail. Through Part B, an unsupervised K-Means clustering approach effectively mapped over 4,300 online retail customers into High-Value, At-Risk, and Average segments using RFM metrics. Together, these methodologies supply evidence-based tools for robust inventory management and highly targeted marketing.

2. References

- Walmart Store Sales Dataset ('Walmart (1).csv')
- Online Retail Transaction Dataset ('OnlineRetail (3).csv')
- Scikit-Learn Official Documentation for ensemble and cluster algorithms.
- Python Pandas, NumPy, and Matplotlib documentation for data processing and visualization.
- Intellipaat Capstone Project Submission Guidelines.

3. Appendices

- Code & Analysis: Extensively detailed in 'Capstone_Project.ipynb'.
- Saved Models: 'sales_forecasting_rf_model.pkl' (Random Forest) and 'customer_segmentation_kmeans_model.pkl' (K-Means).
- Forecasting Output: The 12-week store-wise predictions are fully documented in 'walmart_12_week_forecast.csv'.